Maxim Polyakov
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Education

2023 – Candidate of Technical Sciences (PhD equivalent), Specialisation: Mathematical Modelling, Numerical Methods and Software Systems, Volgograd State University, Dissertation title: Mathematical Modelling of Heat-Transfer Dynamics in Multicomponent Biological Tissues: Analysis of Spatial Distributions of Thermodynamic and Brightness Temperatures.
2017 – Master's degree, major in Software Engineering.
2015 – Bachelor's degree, major in Software Engineering.

Academic Appointments

2012-present – Volgograd State University (VolSU).
Current appointments:

- Associate Professor, Department of Information Systems and Computer Modelling;
- Senior Research Fellow, Department of Information Systems and Computer Modelling.

Research Interests

Mathematical and computational modelling of biological systems.
Computational oncology and tumour dynamics.
Data mining and machine learning.

Publication Metrics

Scopus: 26 indexed publications; h-index: 7; Scopus Author ID: 57196275466.
Web of Science: 12 indexed publications; h-index: 4; ResearcherID: D-9107-2018

Research Publications

Publications Indexed in Web of Science and/or Scopus

Journal Articles

1. Polyakov M. A Hybrid Machine Learning Framework for Mechanistically Interpretable Latent Parameter Inference in a Spatiotemporal CAR-T Therapy Model for Solid Tumours. Technologies, 2026, 14, 276.
2. Polyakov M.V., and Sirotin D.S. Numerical Analysis of Antenna Parameter Influence on Brightness Temperature in Medical Microwave Radiometers. Fluctuation and Noise Letters, 2026.
3. Polyakov M. Spatiotemporal Modelling of CAR-T Cell Therapy in Solid Tumours: Mechanisms of Antigen Escape and Immunosuppression. Computation 2026, 14, 87.
4. Polyakov M.V., and Popov I.E. Computer modeling of bioheat transfer for the analysis of brightness temperature distributions. International Journal of Modern Physics C, 2026.
5. Polyakov M.V., Tuchina E.I. Reaction-Diffusion Model of CAR-T Cell Therapy in Solid Tumours with Antigen Escape. Computation 2026, 14, 3.
6. Polyakov M.V., and Sirotin D.S. A Realistic Breast Phantom for Investigating the Features of the Microwave Radiometry Method Using Mathematical and Physical Modelling. Technologies 2025, 13, 106.
7. Polyakov M.V., and Ten V.V. Simulation tumor growth in heterogeneous medium based on diffusion equation. International Journal of Modern Physics C, 2024, 35(1), 2450010.
8. Polyakov M.V., Khoperskov A.V., Agasieva S.V., and Porokhov I.O. Assessment of the Spatial Resolution and Accuracy of Temperature Measurements in Radiothermometry of the Breast. Biomedical Engineering, 2023, Vol. 57, No. 3, 180-184.
9. Klikunova A.Y., Polyakov M.V., Khrapov S.S., and Khoperskov A.V. Problem of Building High-Quality Predictive Model of River Hydrology: The Combined Use of Hydrodynamic Simulations and Intelligent Computing. Communications in Computer and Information Science, 2023, 1909, pp. 191-205.
10. Khoperskov A.V., and Polyakov M.V. Improving the Efficiency of Oncological Diagnosis of the Breast Based on the Combined Use of Simulation Modeling and Artificial Intelligence Algorithms. Algorithms, 2022, 15(8), 292.
11. Polyakov M.V. Computational modeling to determine the physical characteristics of biological tissues for medical diagnosis. International Journal of Engineering Systems Modelling and Simulation, 2020, 11(4), pp. 214-221.
12. Levshinskii V., Polyakov M., Losev A., and Khoperskov A. Verification and Validation of Computer Models for Diagnosing Breast Cancer Based on Machine Learning for Medical Data Analysis. Communications in Computer and Information Science, 2019, v.1084, pp. 447-460.
13. Polyakov M., Khoperskov A., and Zamechnic T. Numerical modeling of the internal temperature in the mammary gland. Lecture Notes in Computer Science, 2017, Vol. 10594, pp. 128-135.

Conference Proceedings

14. Polyakov M. Integration of Machine Learning to Personalise a Mathematical Model of Tumour Dynamics Based on Reaction-Diffusion Equations. 7th International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency (SUMMA), Lipetsk, Russian Federation, 2025, pp. 776-779.

15. Sirotin D., and Polyakov M. Development of Software for Controlling an Anatomical Breast Phantom in Physical Modelling. 7th International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency (SUMMA), Lipetsk, Russian Federation, 2025, pp. 505-508.
16. Kuksa V., and Polyakov M. Developing and Applying a Neural Network System for Text Plagiarism Detection in Higher Education. 4th International Conference on Technology Enhanced Learning in Higher Education (TELE), Lipetsk, Russian Federation, 2024, pp. 412-416.
17. Emelyanov E., and Polyakov M. Automated Hardware and Software Intelligent System for Infrared Thermometry. Proceedings - 2023 5th International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency, SUMMA 2023, 2023, pp. 328-333.
18. Zhitenko M., and Polyakov M. Automated Control System of a Modular Robot Manipulator for Sorting Objects Based on Neural Networks. Proceedings - 2023 5th International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency, SUMMA 2023, 2023, pp. 363-367.
19. Zhitenko M., and Polyakov M. Development of a Robot Arm with Neural Network Control. Proceedings - 2023 3rd International Conference on Technology Enhanced Learning in Higher Education, TELE 2023, 2023, pp. 23-27.
20. Polyakov M., and Khoperskov A. Create Combined Thermometric Datasets for Machine Learning in Medicine. Proceedings - 2022 4th International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency, SUMMA 2022, 2022, pp. 310-315.
21. Sirotin D., and Polyakov M. Hardware and Software Development of the System for Photogrammetry of Small Objects. Proceedings - 2022 2nd International Conference on Technology Enhanced Learning in Higher Education, TELE 2022, 2022, pp. 78-82.
22. Glazunov V.A., and Polyakov M.V. Testing the algorithm of tumor localization in breast cancer on the results of temperature field simulation. Journal of Physics: Conference Series, 2021, 1794(1), 012006.
23. Polyakov M.V. Cluster Analysis of the Results of Numerical Simulation of Thermometric Data. Proceedings of ITNT 2021 – 7th IEEE International Conference on Information Technology and Nanotechnology, IEEE Xplore, 2021, pp. 1-4.
24. Emelyanov E.A., and Polyakov M.V. Neural Network Software to Process Medical Data Modeling. IEEE: 3rd International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency (SUMMA), IEEE Xplore, 2021, pp. 580-585.
25. Polyakov M., Khoperskov A., Borisovskii E., and Emelyanov E. The using of machine learning and neural networks in the processing of computer simulation results for medical diagnostics. CEUR Workshop Proceedings, 2020, Vol. 2667, pp. 189-192.
26. Polyakov M., Levshinskii V., and Khoperskov A. Modeling of brightness temperature in biological tissue. Journal of Physics: Conference Series, 2019, Vol. 1368, 042057.

Other Peer-Reviewed Publications

27. Polyakov M.V., Khoperskov A.V., Agasieva S.V., and Porokhov I.O. Evaluation of spatial resolution and temperature accuracy in breast radiothermometry. Medical technology, 2023, №3, pp. 18-21.
28. Polyakov M.V., Khoperskov, A. V., Gudkov, A. G., and Chizhikov, S. V. Mathematical Modeling of Brightness Temperature in Biological Tissues for Medical Problems. Nanotechnology: Development and Applications – XXI Century, 2023, Vol. 15, № 2, pp. 5-21.
29. Polyakov M.V., Popov I.E., Losev A.G., and Khoperskov A.V. Application of computer simulation results and machine learning in analysis of microwave radiothermometry data. Mathematical Physics and Computer Simulation, 2021, Vol. 24, № 2, pp. 27-37.
30. Polyakov M.V., Khoperskov A.V., and Svetlov A.V. Simulation modeling of radiation field in biological tissue of mammary gland. Journal of Computational and Engineering Mathematics, 2017, Vol. 4, №. 3, pp. 3-10.
31. Polyakov M.V., and Khoperskov, A. V. Computational Experiments for the Study of Radiation and Thermal Fields in Biological Tissues. Engineering Journal of Don, 2017, No. 2, Article 4239.
32. Polyakov M.V., and Khoperskov, A. V. Mathematical Modeling of Radiation Fields in Biological Tissues: The Definition of the Brightness Temperature for the Diagnosis. Science Journal of Volgograd State University. Mathematics. Physics, 2016, No. 5 (36), pp. 73-84.

Research Grants and Projects

Principal Investigator

1. Russian Science Foundation (RSF), Grant No. 25-71-00041, Mathematical Modelling of the Spatiotemporal Dynamics of Solid Tumours under CAR T Cell Therapy, 2025-2027.
2. Russian Foundation for Basic Research (RFBR), Grant No. 19-47-343008, Development of a Mathematical Modelling Method for Thermal Processes in the Breast for Diagnostic Applications, 2019-2021.

Research Team Member

3. State Assignment of the Ministry of Science and Higher Education of the Russian Federation, No. FZUU-2026-0005, Development of Methods for Modelling Complex Systems through the Integration of Direct Computational Experiments and Data Mining, 2026–present.
4. Russian Science Foundation (RSF), Grant No. 25-21-00330, Development of Methods for Improving the Quality of Training Datasets for the Intelligent Analysis of Thermometric Data, 2025.
5. Russian Science Foundation (RSF), Grant No. 23-71-00016, Mathematical Modelling of Continuum Dynamics: Development of Parallel Algorithms and Software for GPU-Based Supercomputers, 2023-2026.
6. State Assignment of the Ministry of Science and Higher Education of the Russian Federation, No. 0633-2020-0003, project Development of Parallel Algorithms and Software for Modelling the Nonlinear Dynamics of Continuous Media, 2020-2022.
7. Russian Foundation for Basic Research (RFBR), Grant No. 19-37-90142, Computer Modelling of Physical Processes within the Breast: Model Verification and Improving the Effectiveness of Medical Diagnostics, 2019-2021.
8. Russian Foundation for Basic Research (RFBR), Grant No. 19-01-00358, Mathematical Models of Radiation Fields and Analysis of Microwave Radiothermometry Data, 2019-2021.
9. State Assignment of the Ministry of Education and Science of the Russian Federation, No. 2.852.2017/4.6, project Development of Software for Modelling Physical Media and Natural Phenomena, 2017-2019.
10. Russian Foundation for Basic Research (RFBR), Grant No. 15-47-02642, Computational Experiments Investigating the Characteristics of Spatiotemporal Temperature Distributions in Breast Tissue, 2015-2017.

Conference Presentations

1. 26th All-Russian Conference of Young Scientists on Mathematical Modelling and Information Technologies, Novosibirsk, 2025.
2. 7th International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency, Lipetsk, 2025.
3. 4th International Conference on Technology Enhanced Learning in Higher Education (TELE 2024), Lipetsk, 2024.
4. 5th International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency (SUMMA 2023), Lipetsk, 2023.
5. 3rd International Conference on Technology Enhanced Learning in Higher Education (TELE 2023), Lipetsk, 2023.
6. 4th International Conference on Control Systems, Mathematical Modeling, Automation and Energy Efficiency (SUMMA 2022), Lipetsk, 2022.
7. 2nd International Conference on Technology Enhanced Learning in Higher Education (TELE 2022), Lipetsk, 2022.
8. International Conference PhysicA.SPb/2021, St Petersburg, 2021.
9. 7th International Conference and Youth School on Information Technology and Nanotechnology, Samara, 2021.
10. 8th International Conference on Mathematical Biology and Bioinformatics, Pushchino, 2020.
11. 14th Masherov Readings, Vitebsk State University, Vitebsk, 2020.
12. 6th International Conference and Youth School on Information Technology and Nanotechnology, Samara, 2020.
13. 3rd International Conference on Creativity in Intelligent Technologies and Data Science (CIT&DS 2019), Volgograd, 2019.
14. 5th International Conference and Youth School on Information Technology and Nanotechnology, Samara, 2019.
15. 7th International Conference on Mathematical Biology and Bioinformatics, Pushchino, 2018.
16. 6th International Conference on Health Information Science (HIS 2017), Moscow, 2017.
17. 24th International Scientific Conference for Undergraduate and Postgraduate Students and Young Scientists “Lomonosov”, Moscow, 2017.
18. 13th All-Russian School and Conference of Young Scientists on Large-Scale Systems Control, Samara, 2016.
19. 3rd International Conference on Geometric Analysis and Its Applications, Volgograd, 2016.
20. 12th All-Russian School and Conference of Young Scientists on Large-Scale Systems Control, Volgograd, 2015.

Courses Taught

1. Fundamentals of Modelling.
2. Methods of Mathematical Modelling.
3. Data Structures and Algorithms.

Peer Review for Academic Journals

Computer Methods and Programs in Biomedicine, International Journal of Molecular Sciences, International Journal of Thermal Sciences, Algorithms, Sensors, Diagnostics, Processes, Mathematics, Computers, Sustainability, Applied Sciences, etc.